CityViews: Benchmarking Satellite and Street-Level Imagery in Urban Vision-Language Models

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KOÇ ÜNİVERSİTESİ

Does a second urban view help a vision-language model?

Only when the answer lives *between* the views.

Fusion synergy = accuracy with both views − accuracy of the better single view, averaged over tasks.
Same 9B model, same questions, same protocol; the two task regimes give opposite answers:

+1.6 pts
urban-attribute tasks (7)
land use, road type, building height, density, …

+46.6 pts
cross-view tasks (5)
camera direction, satellite ↔ street matching

1 The problem

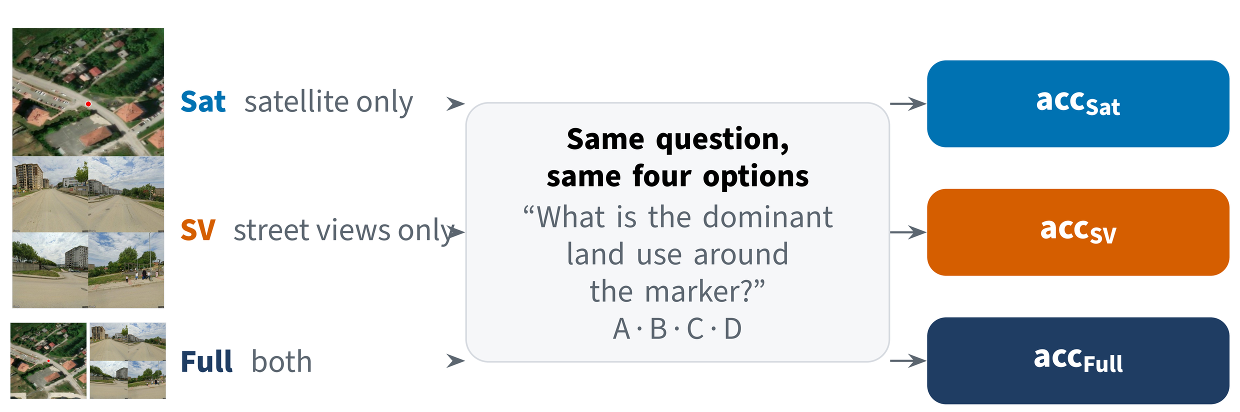
- Satellite tiles and street-level photos are *assumed* to be complementary for urban VLMs, but this is rarely tested directly.
- One aggregate accuracy conflates two very different situations: a question one view could answer alone, and a question that genuinely needs both.
- Street-level coverage is expensive to acquire, license and serve; its *marginal* value should be measured *before* deployment, not after.

2 CityViews at a glance

40 cities	**32** countries, 6 continents	**12** tasks, 2 regimes
34 484 questions	**4 168** locations	**0** human annotations

Each location = one **marked satellite tile** (Esri / NAIP / IGN, sub-metre) + **four road-aligned street views**. Labels from OpenStreetMap tags and geometry, questions from hand-written templates: **no LLM or VLM anywhere in the pipeline**, so CityViews regenerates deterministically from source.

3 The protocol: one question, three inputs

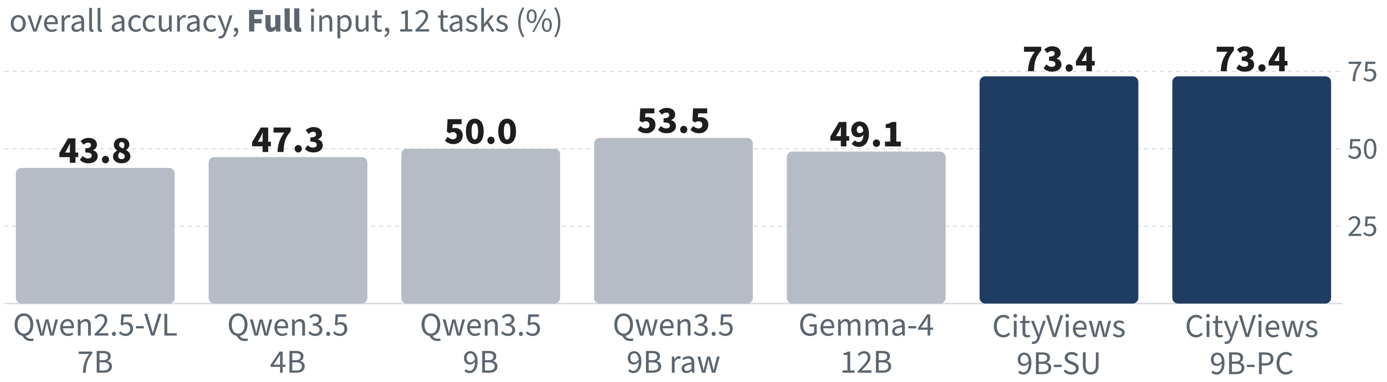


synergy = $acc_{Full} - \max(acc_{Sat}, acc_{SV})$

Synergy within ±3 pts is read as "no meaningful benefit from the second view" (equivalence band). 4 734 held-out questions; identical text across conditions, so any difference is due to the images.

4 Does the benchmark separate models?

overall accuracy, **Full** input, 12 tasks (%)

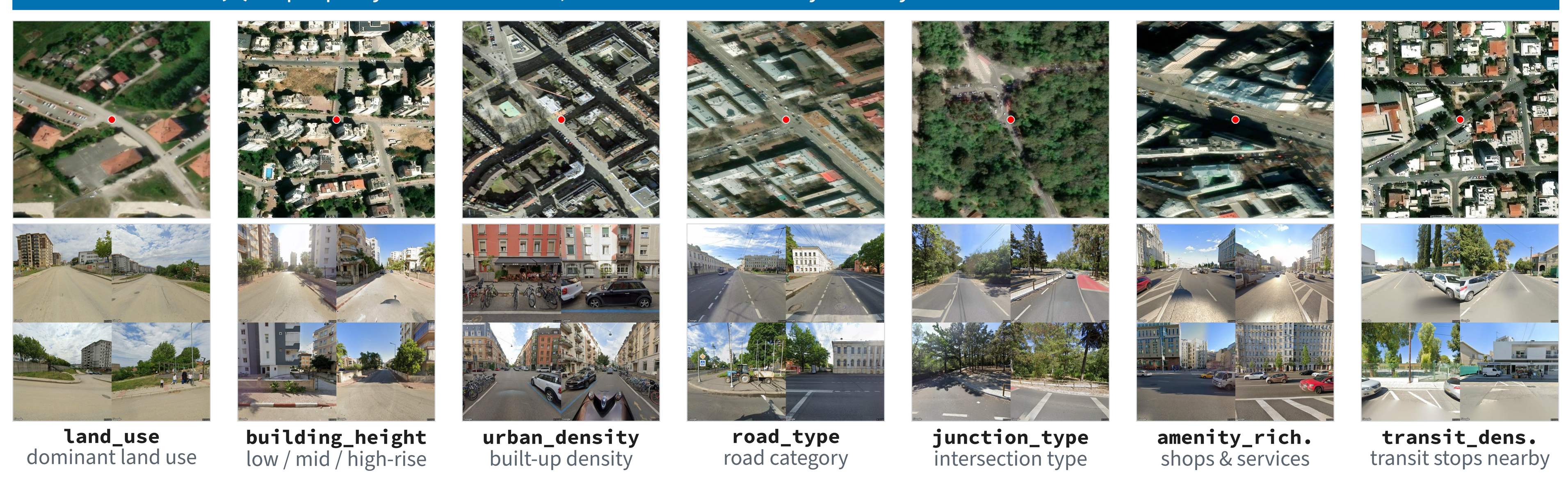


Zero-shot generalist VLMs land at 44–54 %; task-tuned CityViews-9B reaches 73.4 % and still does not saturate (amenity richness < 60 %, camera direction 44 %).

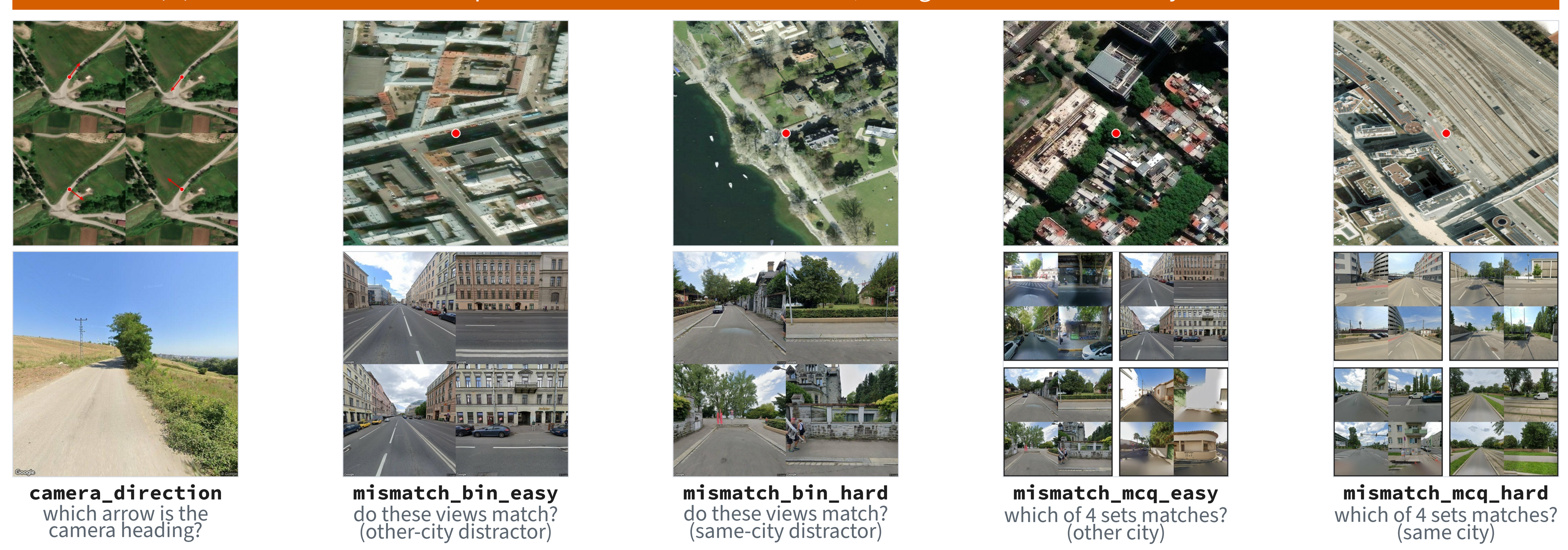
5 Twelve tasks, two regimes

Each cell: marked satellite tile (top) and street view(s) (bottom) for one location. Four-way multiple choice unless noted.

Urban-attribute (7) a property of one location; either view alone may already suffice

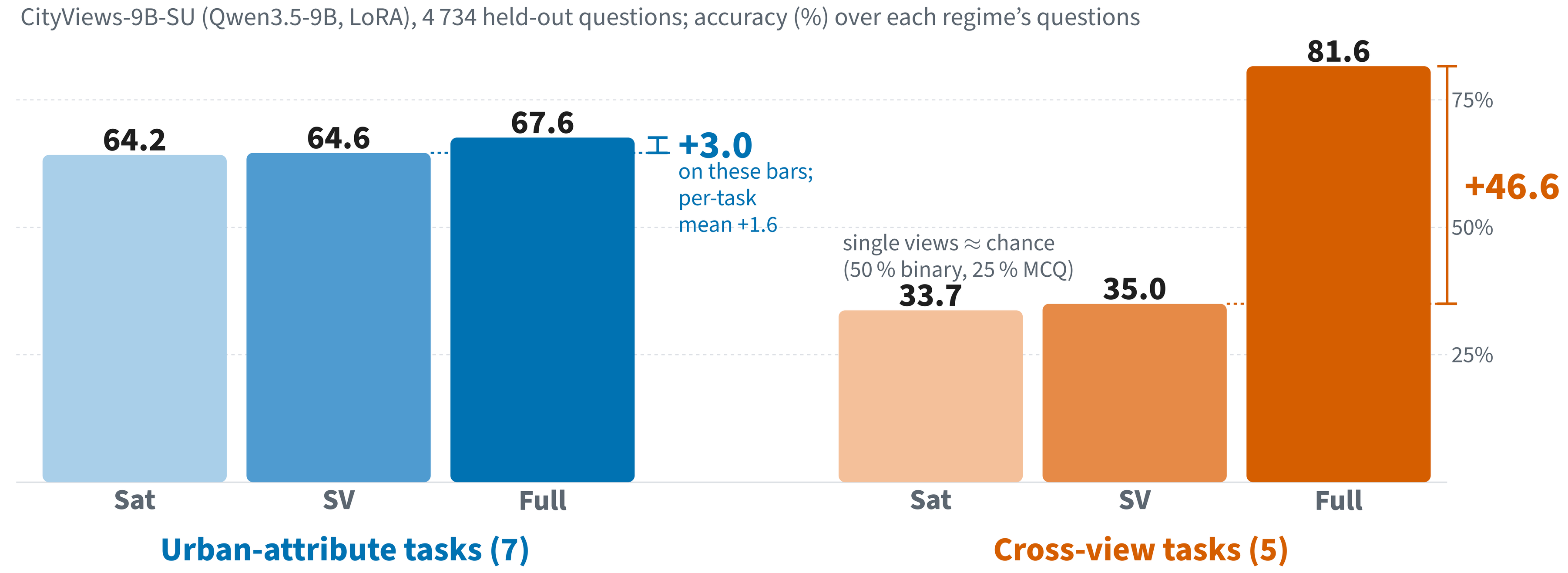


Cross-view (5) the answer is the correspondence between the two views; a single view is at chance by construction



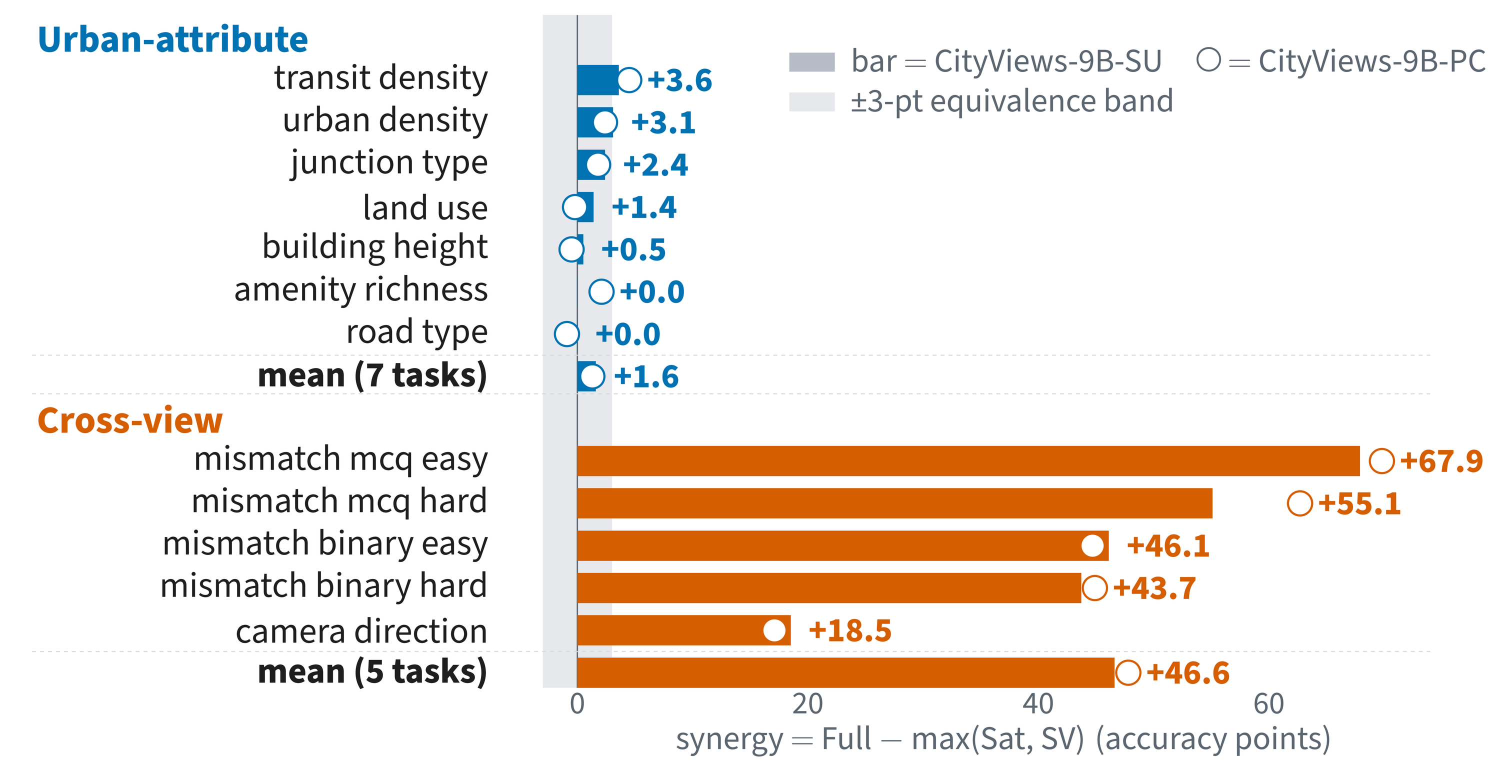
6 Headline: the same model, opposite answers

CityViews-9B-SU (Qwen3.5-9B, LoRA), 4 734 held-out questions; accuracy (%) over each regime's questions



On urban attributes the two views are largely **redundant**: either alone reaches ≈ 64 %, and both together add +3.0 on pooled accuracy, +1.6 averaged per task (panel 7), both inside the ±3 equivalence band. On cross-view tasks the same model goes from chance to 81.6 % only when it holds both views. **Whether the second view helps is a property of the task, not of the model.**

7 Per-task fusion synergy



8 The small urban gain is not an artefact

−0.9 pts
…of fine-tuning
raw zero-shot Qwen3.5-9B: urban synergy −0.9, cross-view still +22.9

+0.5 pts
…of the view budget
train with one forward street view instead of four: Full 65.3 vs 67.6, synergy +0.5

86.9 %
…of inference-time views
Qwen3.6-27B: pruning to the forward view keeps 86.9 % of correct answers; no street view at all keeps 61.7 %

9 Three models, one protocol

Model	Tasks	Sat	SV	Full	Synergy
CityViews-9B-SU	urban (7)	64.2	64.6	67.6	**+1.6**
	cross-view (5)	33.7	35.0	81.6	**+46.6**
CityViews-9B-PC	urban (7)	63.9	64.6	66.8	**+1.3**
	cross-view (5)	34.3	33.7	82.5	**+47.8**
Qwen3.5-9B, raw	urban (7)	41.3	49.9	49.5	**−0.9**
	cross-view (5)	35.5	34.4	59.1	**+22.9**

SU: seen/unseen split (8 held-out cities); PC: per-city split. Accuracy (%) averaged over tasks; Synergy = mean of per-task synergies.

10 Takeaways

- Fusion is **decisive when the answer is the relationship between views**, and near-zero when either view already carries the evidence.
- Aggregate both-views accuracy hides this: **report matched single-view baselines.**
- **Measure before you collect**: for attribute-style questions, street-level imagery may not pay for itself.

paper · code · data
ezelm.github.io/cityviews

Scope. Behavioural synergy of current VLMs on multiple-choice questions with OSM-certified labels, Europe-weighted; not an information-theoretic claim about the views.